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国家自然科学基金
资助项目结题/成果报告

资助类别： 青年科学基金项目（C类）[原青年科学基金项目]

亚类说明：

附注说明：

项目名称： 基于分子动力学模拟与机器学习的高性能聚芳醚材料基因工程研究

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项目摘要

中文摘要:

针对“经验驱动、实验试错”传统材料研发模式效率低下的问题，基于航空航天、武器装备领域对兼具高耐热等级、优异力学性能及优良加工性能聚芳醚树脂的迫切需求，本项目在材料基因工程思想的引领下，结合分子动力学模拟与机器学习技术，开展“数据驱动、理论预测、实验验证”新模式的聚芳醚树脂研究工作。通过建立可复现聚芳醚树脂实验测试结果的模拟计算方法，深入探究聚芳醚树脂分子结构对其耐热性、力学性能和加工性能的影响机制。利用模拟计算结果建立的聚芳醚树脂理论计算数据库，设计聚芳醚树脂新型多层次材料特征，构建可合理预测聚芳醚树脂上述性能的高精度机器学习模型，提出可同步提升聚芳醚树脂耐热性、力学性能和加工性能的分子结构设计策略，解决聚芳醚树脂高耐热等级、优异力学性能与优良加工性能难以兼得的难题。利用实验合成与表征工作，检验理论预测结果的可靠性，研制出综合性能优越的新型聚芳醚树脂。

Abstract:

To overcome the bottlenecks of low efficiencies in traditional materials research and development, and to satisfy urgent demands for poly(aryl ether)s with high thermo-resistance, excellent mechanical properties and good processability in fields of aerospace and weaponry, this project combines molecular dynamics simulations with machine learning technologies to implement innovative development mode for poly(aryl ether)s based on data-driven approaches, theoretical predictions and experimental validations guided by the ideas of materials genome engineering. This project aims to establish high-accuracy simulation methods which can reproduce experimental results of poly(aryl ether) samples. Thus, the relationships between molecular structures of poly(aryl ether)s and the thermo-resistance, mechanical properties and the processability are studied. Furthermore, new hierarchical features of poly(aryl ether)s and high-accuracy machine learning models for predicting the aforementioned properties of poly(aryl ether)s are established based on the database of simulation results. And the molecular design strategy for enhancing the thermo-resistance, mechanical properties and the processability of poly(aryl ether)s can be proposed, and the problem of simultaneously improving the thermo-resistance, mechanical properties and the processability of poly(aryl ether)s will be solved. Additionally, novel poly(aryl ether)s with superior comprehensive properties will be developed with experimental synthesis and characterization efforts which are made to evaluate the reliability of theoretical predictions.

关键词（用分号分开）： 高分子材料计算与模拟； 分子动力学模拟； 机器学习； 材料基因工程； 高性能聚芳醚



Keywords (separated by;): Polymer modelling and simulation; Molecular dynamics simulation; Machine learning; Materials genome engineering; High-performance poly(aryl ether)s

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结题摘要

中文摘要（对项目的背景、主要研究内容、重要结果、关键数据及其科学意义等做简单概述）：

本项目面向航空航天、武器装备领域对兼具高耐热等级、优异力学性能及优良加工性能聚芳醚树脂的迫切需求，针对“经验驱动、实验试错”传统材料研发模式效率低下的瓶颈问题，开展数据驱动的高性能聚芳醚树脂设计研究。

开发了一种可复现聚芳醚树脂实验测试结果的全原子结构模型搭建算法，基于微观模型的结构参数、分子动力学模拟计算结果，以及树脂实验测试数据，建立了目前涵盖聚芳醚数量最多（2800余种聚芳醚）的聚芳醚研发专用数据库，为实现数据驱动的高性能聚芳醚研发新范式，提供了关键物质基础保障。

提出了线形无规共聚物数字化表示方法，可通过自动化计算流程，将聚芳醚局域分子结构特征，以及分子链构象、物化性质等长程信息，快速、高效地转化为模型可读形式，提取出与材料性能高度相关的材料特征，以此建立了高精度、强泛化、可解释的聚芳醚性能预测模型。该数字化表示方法不仅可用于聚芳醚树脂研究，更为线形无规共聚物的数字化处理，提供了一种通用技术。

揭示了影响树脂耐热性、溶解性、力学性能的本征因素，提出了高性能聚芳醚树脂的设计研发策略，成功研制出多个系列综合性能优越的新型聚芳醚树脂。研制出的新型聚芳醚树脂玻璃化转变温度达372 °C，在室温下可溶于多种常用溶剂，且拉伸强度达91 MPa。

本项目发展并实践了“数据驱动、理论预测、实验验证”的高性能聚芳醚树脂研发新模式，为先进高分子材料的快速研发提供新思路。

Abstract (Brief description of research background, main methods, contributions, and research data):

This project conducts a data-driven research on the design of high-performance poly(arylene ether)s (PAEs) to address the urgent demands in fields of aerospace and weaponry for PAEs with high thermo-resistance, excellent mechanical properties and good processability, and to overcome the low efficiency in the traditional materials research and development.

An atomistic modeling algorithm capable of reproducing experimental results for PAEs has been developed. Based on structural parameters obtained from simulated models, molecular dynamics simulation results, and experimental data, a dedicated PAEs database (currently covering more than 2,800 PAEs) has been established. This database provides essential foundations for enabling a data-driven paradigm in high-performance PAE development.

A numerical representation method for linear random copolymers has been proposed. Through an automated computational workflow, local molecular structural features of PAEs, along with long-range information such as chain conformations and physicochemical properties, can be rapidly and efficiently converted into machine-readable formats. This representation approach allows extraction of features that are highly correlated with PAEs properties, enabling the construction of prediction models with high predictive accuracies, generalization abilities, and interpretabilities. This numerical representation method is not only applicable to PAEs research but also offers a general technique for the numerical processing of linear random copolymers.

Design strategies for high-performance PAEs have been proposed based on the unearthed mechanisms governing the thermal resistance, solubility, and mechanical properties of PAEs. Consequently, several new series of PAEs possessing outstanding comprehensive performance have been successfully developed. The newly developed resin exhibits a high glass transition temperature of 372 °C, is soluble in common solvents at room temperature, and possess a tensile strength of 91 MPa.

This project opens a “data-driven, theory-guided, experiment-validated” avenue for the development of high-performance PAEs, offering new ideas for accelerating the development of advanced polymers.



正文

《结题/成果报告》正文分为两个部分：**结题部分和成果部分**。请按照《结题/成果报告》填报说明及撰写要求填写。

(一) 结题部分

1. 研究计划执行情况概述。

(1) 按计划执行情况。

本项目在执行期内，未对研究方案和技术路线进行调整或修改。项目进展顺利，研究目标达成。按计划执行情况如下：

第一年度：利用分子动力学模拟技术，开发了一种线形无规共聚物全原子结构模型搭建算法，并以此建立了聚芳醚树脂微观结构模型，系统研究了聚芳醚树脂单体结构、单体投料比（分子结构）对其耐热性、力学性能和加工性能的影响机制。基于模拟计算结果、微观模型的结构参数，以及聚芳醚树脂各项性能实测数据，建立了聚芳醚树脂研发用数据库，为开展数据驱动的聚芳醚设计研发工作，提供了关键数据支撑。

第二年度：提出了线形无规共聚物数字化表示方法，该方法可高效计算并表征单体分子结构与物化性质、单体投料比、聚合物分子链结构与构象、分子链间相互作用与缠结，以及分子量及其分布情况，通过相关性分析、重要性筛选手段，构建出与聚芳醚树脂耐热性、力学性能及加工性能高度相关的材料特征，为高精度、强泛化机器学习预测模型的建立，提供关键技术保障。

第三年度：分别建立了聚芳醚树脂耐热性机器学习预测模型（单目标优化）、可同时分析树脂耐热性、溶解性、力学性能的预测模型（多目标优化）。揭示了影响树脂综合性能的“基因”，提出了开发兼具高耐热等级、优异力学性能和优良加工性能聚芳醚树脂的分子结构设计策略。在设计研发策略的指导下，设计并制备出综合性能优越的新型聚芳醚树脂，验证了机器学习模型预测结果的合理性与可靠性。

(2) 研究目标完成情况。

本项目预期建立高精度模拟计算方法，揭示聚芳醚树脂分子结构对其综合性